

SYDNEY GRAMMAR SCHOOL



2014 Annual Examination

FORM V

MATHEMATICS EXTENSION 1

Monday 1st September 2014

General Instructions

- Writing time — 2 Hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 100 Marks

- All questions may be attempted.

Section I — 9 Marks

- Questions 1–9 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II — 91 Marks

- Questions 10–16 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your name, class and master on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Ten.
- Write your name and master on this question paper and submit it with your answers.

5A: BDD

5B: MLS

5C: LYL

5D: LRP

5E: PKH

5F: BR

5G: SG

Checklist

- SGS booklets — 7 per boy
- Multiple choice answer sheet
- Candidature — 131 boys

Examiner

BDD

SECTION I - Multiple Choice

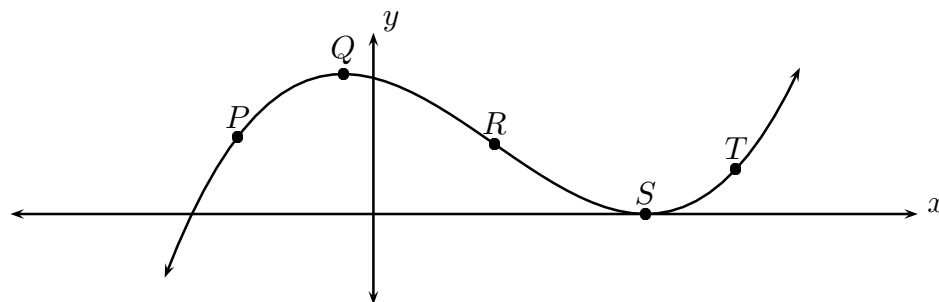
Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Which of the following is a correct expression for a primitive of e^{5x} ?

- (A) $\frac{1}{5}e^{5x}$
- (B) $5e^{5x}$
- (C) $\frac{1}{6}e^{6x}$
- (D) $\frac{1}{5x+1}e^{5x+1}$

QUESTION TWO



The graph of a function is sketched above. In this graph, Q and S are stationary points and R is a point of inflexion.

At which of the marked points is the second derivative $y'' > 0$?

- (A) P and Q
- (B) S and T
- (C) R , S and T
- (D) P and T

Exam continues next page ...

QUESTION THREE

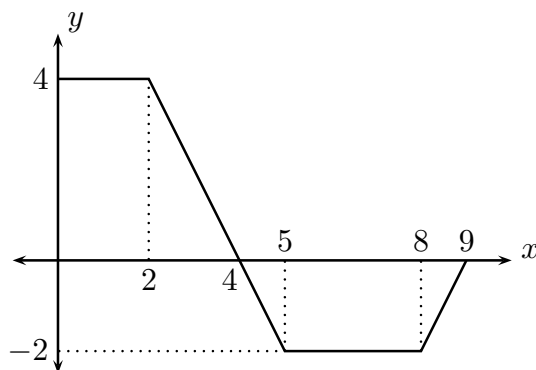
Which of the following integration statements is correct?

(A) $\int (x^2 + 1)^2 dx = \frac{(x^2 + 1)^3}{6x} + C$

(B) $\int \ln x dx = \frac{1}{x} + C$

(C) $\int \frac{3}{x^2} dx = -\frac{1}{x^3} + C$

(D) $\int \frac{2x + 6x^2}{x} dx = 2x + 3x^2 + C$

QUESTION FOURThe function $y = f(x)$ is sketched above. The correct value of $\int_0^5 f(x) dx$ is:

(A) 4

(B) 11

(C) 13

(D) 20

QUESTION FIVEWhich statement is true of the quadratic $y = 4x^2 + 24x + 36$?

(A) It is positive definite;

(B) It has two unreal zeroes;

(C) It is a perfect square;

(D) The zeroes add to 6.

Exam continues overleaf ...

QUESTION SIX

The correct solution of $\frac{x}{x-3} > 0$ is:

- (A) $x < 0$ or $x > 3$
- (B) $0 < x < 3$
- (C) $x > 0$
- (D) $x > 0$ or $x > 3$

QUESTION SEVEN

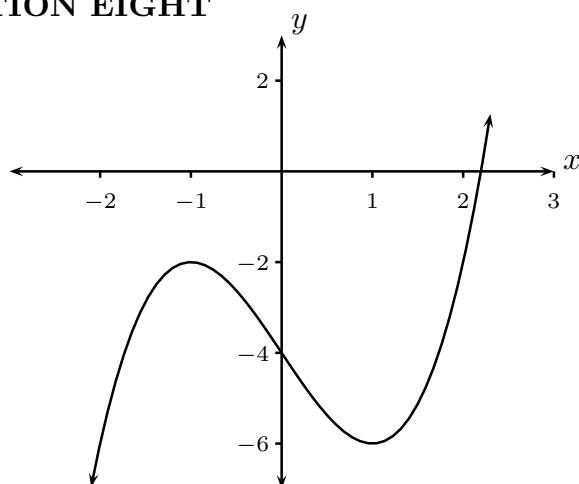
A function $y = f(x)$ is evaluated at points in the interval $0 \leq x \leq 4$, as in the table below.

x	0	1	2	3	4
$f(x)$	2	4	3	5	4

An estimate of $\int_0^4 f(x) dx$ using two applications of Simpson's rule is:

- (A) 5
- (B) 8
- (C) 16
- (D) 24

QUESTION EIGHT



The graph of $y = x^3 - 3x - 4$ is sketched above. What is the smallest value of a constant c such that $y = c$ intersects the graph of the cubic at least twice?

- (A) -2
- (B) -4
- (C) -6
- (D) -7

QUESTION NINE

Which of the following functions does not have a horizontal asymptote $y = 1$?

- (A) $y = 1 + e^x$
- (B) $y = \frac{x^2 + 1}{x^2 - 1}$
- (C) $y = 3 - \frac{2x + 1}{x + 1}$
- (D) $y = \frac{3x^2 + 1}{3x + 1}$

_____ End of Section I _____

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION TEN (13 marks) Use a separate writing booklet.	Marks
(a) Find the exact value of $x^3 + x^2 - 3x + 1$ when $x = -\sqrt{2}$.	2
(b) Solve $x - \frac{4}{x-1} = 1$.	2
(c) Find the exact solution of $\log_e(2x - 4) = 3$.	2
(d) Differentiate:	
(i) $y = \ln(3x + 1)$	1
(ii) $y = 5xe^x$	2
(e) For what values of x is the function $y = x^2 - 6x + 3$ decreasing?	2
(f) Simplify $\frac{a+b}{\frac{1}{a} + \frac{1}{b}}$.	1
(g) Solve $\tan 2\theta = 0$, for $0^\circ \leq \theta \leq 360^\circ$.	1

QUESTION ELEVEN (13 marks) Use a separate writing booklet.**Marks**

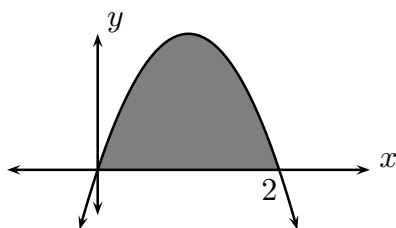
(a) Find:

(i) $\int \frac{2}{x} dx$ 1

(ii) $\int 4\sqrt{x} dx$ 1

(iii) $\int \frac{2x}{x^2 + 1} dx$ 1

(b)

Find the area bounded by the curve $y = 6x - 3x^2$ and the x -axis, as shaded above.2(c) Prove that $y = \frac{3}{x^2}$ is concave up for all $x \neq 0$.2(d) Solve $2\sin^2 \theta - \sin \theta - 1 = 0$, for $0^\circ \leq \theta \leq 360^\circ$.3(e) Find all values of a for which $\int_1^a (x+1) dx = 6$.3**QUESTION TWELVE** (13 marks) Use a separate writing booklet.**Marks**(a) A certain function has derivative $y' = 8x^3 - 6x^2 + 4x - 2$ and passes through the point $(2, 9)$. Find an expression for the function y .3(b) Find constants a , b and c such that $2x^2 - 3x + 5 \equiv a(x-1)^2 + b(x-1) + c$.3(c) Consider the quadratic $(k+3)x^2 + 2kx + 4$, where k is a constant.

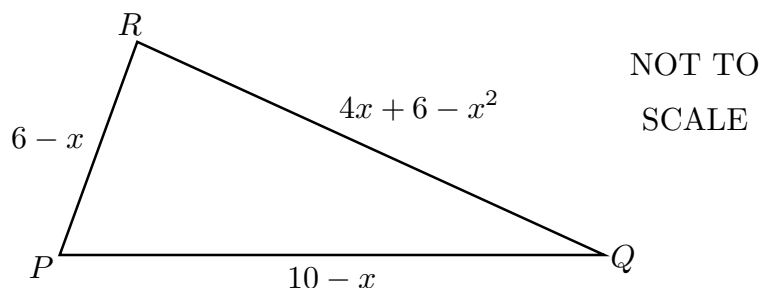
(i) Find a simplified expression for its discriminant.

1(ii) For what values of k does the quadratic have no real zeroes?2(iii) Explain why there are no values of k for which the quadratic is negative definite.1(d) The quadratic equation $2x^2 - 14x + c = 0$, where c is a constant, is known to have roots that differ by 3. By letting the two roots be α and $\alpha + 3$, find the value of c .3**Exam continues overleaf ...**

QUESTION THIRTEEN (13 marks) Use a separate writing booklet.

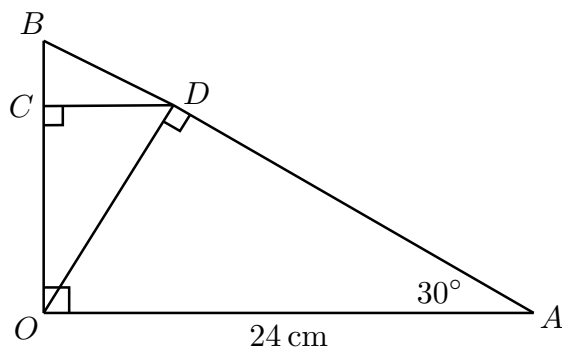
Marks

- (a) The length of the three sides of $\triangle PQR$ pictured below are known in terms of x , where $0 \leq x \leq 3$.



- (i) Write down a simplified expression for the perimeter of the triangle. 1
- (ii) Use calculus to find the **maximum** possible value of the perimeter as x varies. 3
- (iii) What is the **minimum** value of the perimeter? 2

(b)



In the diagram above, $OA = 24$ cm and $\angle BAO = 30^\circ$.
Also $\angle BOA = \angle ODA = \angle OCD = 90^\circ$.

Find the exact length of BC .

3

- (c) (i) Shade the region bounded by $y = x^2 + 1$, the coordinate axes and the line $x = 2$. 1
- (ii) The region in part (i) is rotated about the x -axis to form a solid of revolution. 3
Calculate the volume of this solid.

QUESTION FOURTEEN (13 marks) Use a separate writing booklet.**Marks**(a) Consider the function $f(x) = x^2(16 - x^2)^8$.(i) Find the derivative $f'(x)$.**2**(ii) Factorise your answer in part (i) and hence find the x -coordinates of all stationary points of the function $y = f(x)$.**2**

(b) The country of Pecunia is suffering from an extremely high rate of inflation. A certain family buys a loaf of bread at the start of each week. Over the course of a fifty-two week period, the price of a loaf of bread at their bakery increased by 2% each week. The cost of the loaf at the beginning of the first week was \$2.

Let T_n be the price of the loaf at the beginning of week n and assume it is modelled by a geometric sequence.(i) Write down an expression for T_n using the information given.**1**

(ii) What is the cost of their last loaf, bought in the fifty-second week?

1

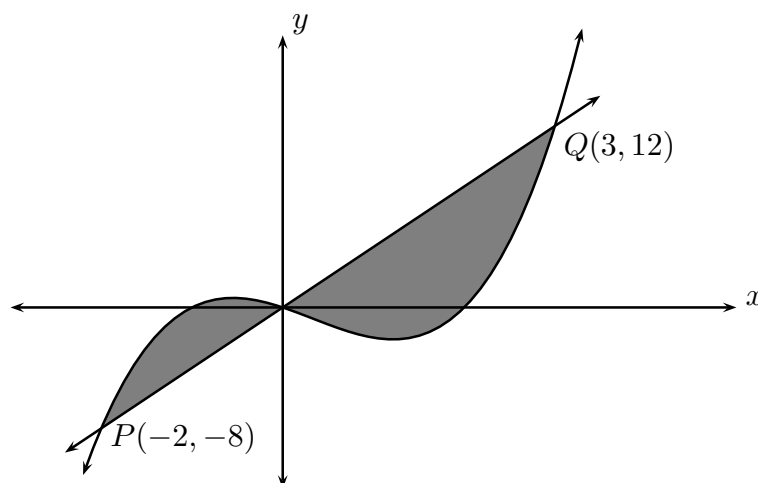
(iii) In what week does their loaf first cost them over \$4?

2

(iv) How much does the family spend on bread over the fifty-two week period?

1

(c)



The graphs of $y = x^3 - x^2 - 2x$ and $y = 4x$ are sketched above. The graphs intersect at $P(-2, -8)$, $Q(3, 12)$ and the origin $O(0, 0)$. (You need NOT show this).

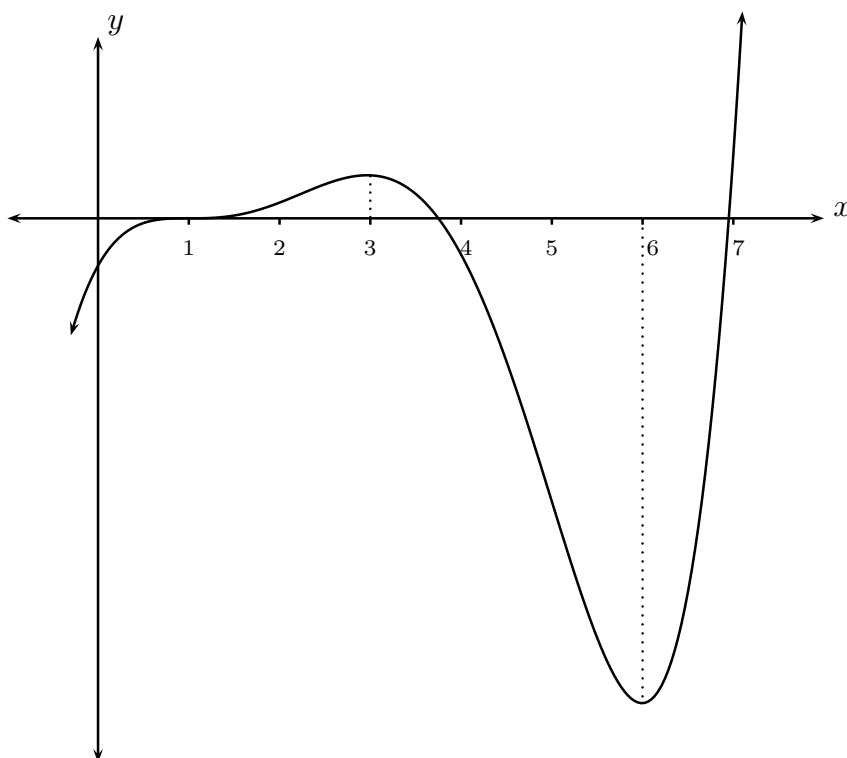
Find the area of the region bounded by the cubic and the line, which is shaded in the diagram.

4

QUESTION FIFTEEN (13 marks) Use a separate writing booklet.

Marks

(a)



The graph above is stationary when $x = 1$, $x = 3$ and $x = 6$ and has points of inflexion when $x = 1$, $x = 2$ and $x = 5$. Sketch a possible graph of its derivative. Be sure to label the x -axis in your solution with the integer values from $x = 1$ to $x = 7$.

3

(b) Consider the curve $y = \frac{x+1}{x^2}$.

(i) Write down any intercepts with the coordinate axes.

1

(ii) Write down the equation of the vertical asymptote.

1

(iii) With working to justify your answer, find the equation of the horizontal asymptote.

1

(iv) Find the derivative y' .

1

You may assume that the second derivative is $y'' = \frac{2x+6}{x^4}$.

(v) Find any stationary points and determine their nature.

2

(vi) Find any points of inflexion.

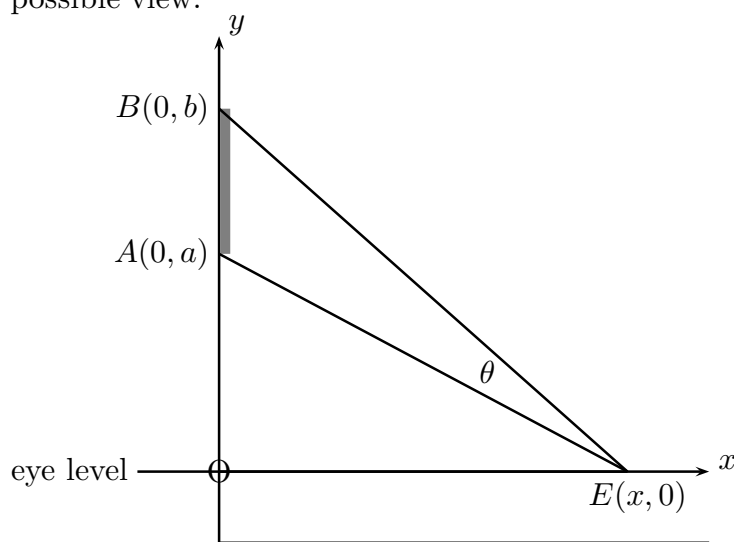
2

(vii) Sketch the curve, showing clearly the information found above.

2

QUESTION SIXTEEN (13 marks) Use a separate writing booklet.**Marks**

- (a) Use the discriminant to find any tangents to the curve $y = x^3 + 2x^2 - 3x$ that pass through the origin. Hint: Let $y = mx$. **3**
- (b) (i) Differentiate $x^3 \ln x$. **1**
- (ii) Hence find $\int x^2 \ln x \, dx$. **1**
- (iii) Given the definition $\int_0^1 x^2 \ln x \, dx = \lim_{\epsilon \rightarrow 0} \int_{\epsilon}^1 x^2 \ln x \, dx$, evaluate $\int_0^1 x^2 \ln x \, dx$. **1**
- (c) Staff at an art gallery wish to ensure that patrons viewing the paintings have the best possible view.



Suppose a painting is hung on the wall, with the top and bottom of the painting at heights b and a metres respectively above eye level. Suppose the painting subtends an angle θ , where $0^\circ \leq \theta \leq 90^\circ$, at the eye of the viewer standing at E , x metres from the wall.

- (i) Use the cosine rule in $\triangle ABE$ to show that **2**

$$\cos \theta = \frac{x^2 + ab}{\sqrt{a^2 + x^2} \sqrt{b^2 + x^2}}.$$

- (ii) Use the identity $\tan^2 \theta = \frac{1 - \cos^2 \theta}{\cos^2 \theta}$ to show that **2**

$$\tan \theta = \frac{(b - a)x}{x^2 + ab}.$$

- (iii) Use calculus to find the value of x that maximises $\tan \theta$, and hence the position that maximises the observer's angle of view θ . **3**

————— End of Section II —————

END OF EXAMINATION

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

NAME:

CLASS: MASTER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

SECTION I - Multiple Choice**QUESTION ONE**

The correct is answer is A

QUESTION TWO

$y'' > 0$ where is it concave up, thus the correct answer is B

QUESTION THREE

The correct answer is D

QUESTION FOUR

Note the limits – we are not finding the integral over the whole domain.

$$\begin{aligned}\int_0^5 f(x) dx &= 4 \times 2 + \frac{1}{2} \times 4 \times 8 - \frac{1}{2} \times 2 \times 1 \\ &= 11\end{aligned}$$

The correct answer is B

QUESTION FIVE

The discriminant is $\Delta = 24^2 - 4 \times 4 \times 36 = 0$ so it is NOT positive definite.

Since $\Delta = 0$, it doesn't have 2 distinct zeroes.

$4x^2 + 24x + 36 = (2x + 6)^2$, so it is a perfect square.

The zeroes add to $-24/4 = -6$. The correct answer is C

QUESTION SIX

Multiply by $(x - 3)^2$. Hence $x(x - 3) > 0$. Thus $x < 0$ or $x > 3$. The correct answer is A

QUESTION SEVEN

$$\begin{aligned}\int_0^4 f(x) dx &= \int_0^2 f(x) dx + \int_2^4 f(x) dx \\ &\doteq \frac{1}{6}(2 - 0)(2 + 4 \times 4 + 3) + \frac{1}{6}(4 - 2)(3 + 4 \times 5 + 4) \\ &= \frac{1}{6}(2)(2 + 4 \times 4 + 2 \times 3 + 4 \times 5 + 4) \\ &= 16\end{aligned}$$

The correct answer is C

QUESTION EIGHT

The line $y = -6$ cuts the cubic twice, but a lower horizontal line only cuts once. Hence C.

QUESTION NINE

The correct answer is D.

SECTION II - Written Response**QUESTION TEN**

$$\begin{aligned}
 \text{(a)} \quad x^3 + x^2 - 3x + 1 &= (-\sqrt{2})^3 + (-\sqrt{2})^2 - 3 \times (-\sqrt{2}) + 1 \\
 &= -2\sqrt{2} + 2 + 3\sqrt{2} + 1 \\
 &= \sqrt{2} + 3
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad x - \frac{4}{x-1} &= 1 \\
 \frac{x(x-1)}{x-1} - \frac{4}{x-1} &= 1 \\
 \frac{x^2 - x - 4}{x-1} &= 1 \\
 x^2 - x - 4 &= x - 1 \\
 x^2 - 2x - 3 &= 0 \\
 (x-3)(x+1) &= 0 \\
 \text{Hence } x &= -1 \text{ or } x = 3.
 \end{aligned}$$

$$\text{(c)} \quad x = \frac{1}{2}(4 + e^3).$$

$$\begin{aligned}
 \text{(d)} \quad \text{(i)} \quad y' &= \frac{3}{3x+1} \\
 \text{(ii)} \quad y' &= 5e^x + 5xe^x \\
 &= 5(1+x)e^x
 \end{aligned}$$

(e) It is decreasing when $y' < 0$. Thus:

$$\begin{aligned}
 2x - 6 &< 0 \\
 x &< 3
 \end{aligned}$$

$$\begin{aligned}
 \text{(f)} \quad \frac{a+b}{\frac{1}{a} + \frac{1}{b}} &= \frac{a+b}{\frac{b+a}{ab}} \\
 &= \frac{(a+b)ab}{(b+a)} \\
 &= ab
 \end{aligned}$$

$$\begin{aligned}
 \text{(g)} \quad 2\theta &= 0^\circ, 180^\circ, 360^\circ, 540^\circ, 720^\circ \quad \text{for } 0^\circ \leq 2\theta \leq 720^\circ \\
 \theta &= 0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ
 \end{aligned}$$

QUESTION ELEVEN

$$\begin{aligned}
 \text{(a)} \quad \text{(i)} \quad \int \frac{2}{x} dx &= 2 \log_e x + C \\
 \text{(ii)} \quad \int 4\sqrt{x} dx &= \int 4x^{\frac{1}{2}} dx \\
 &= \frac{8}{3}x^{\frac{3}{2}} + C \\
 &= \frac{8}{3}\sqrt{x^3} + C \\
 \text{(iii)} \quad \int \frac{2x}{x^2+1} dx &= \log_e(x^2+1) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) Area} &= \int_0^2 (6x - 3x^2) dx \\
 &= [3x^2 - x^3]_0^2 \\
 &= (12 - 8) - (0 - 0) \\
 &= 4 \text{ u}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{(c) } y &= 3x^{-2} \\
 y' &= -6x^{-3} \\
 y'' &= 18x^{-4} \\
 y'' &= \frac{18}{x^4}
 \end{aligned}$$

$$> 0 \quad (\text{where } x \neq 0)$$

$$\text{(d) } 2 \sin^2 \theta - \sin \theta - 1 = 0$$

$$(2 \sin \theta + 1)(\sin \theta - 1) = 0$$

$$\text{Hence } \sin \theta = -\frac{1}{2}$$

$$\text{or } \sin \theta = 1$$

$$\theta = 210^\circ, 330^\circ$$

$$\theta = 90^\circ$$

$$\text{(e) } \int_1^a (x+1) dx = 6$$

$$\left[\frac{1}{2}x^2 + x \right]_1^a = 6$$

$$\left(\frac{1}{2}a^2 + a \right) - \left(\frac{1}{2} + 1 \right) = 6$$

$$a^2 + 2a - 3 = 12$$

$$a^2 + 2a - 15 = 0$$

$$(a+5)(a-3) = 0$$

$$\text{Hence } a = -5 \text{ or } a = 3.$$

QUESTION TWELVE

$$\text{(a) } y' = 8x^3 - 6x^2 + 4x - 2$$

$$y = 2x^4 - 2x^3 + 2x^2 - 2x + C,$$

for some constant C . Since it passes through $(2, 9)$;

$$9 = 2(2)^4 - 2(2)^3 + 2(2)^2 - 2(2) + C$$

$$9 = 32 - 16 + 8 - 4 + C$$

$$C = -11$$

The function is $y = 4x^4 - 2x^3 + 2x^2 - 2x - 11$.

$$\text{(b) Equating coefficients of } x^2 \text{ gives } a = 2.$$

Substituting $x = 1$ gives $c = 4$.

Substituting $x = 0$ gives $a - b + c = 5$, so $b = 1$.

$$\text{Thus } 2x^2 - 3x + 5 = 2(x-1)^2 + 1(x-1) + 4.$$

$$\text{(c) (i) } \Delta = (2k)^2 - 4 \times (k+3) \times 4$$

$$\Delta = 4(k^2 - 4k - 12)$$

$$\Delta = 4(k-6)(k+2)$$

(ii) The quadratic has no real roots if $\Delta < 0$ i.e. if $-2 < k < 6$.

(iii) To be negative definite we need $\Delta < 0$ and $(k+3) < 0$, so that the quadratic is concave down. But $\Delta < 0$ is never true for $k < -2$, and it particular it is not true for $k < -3$.

$$\text{(d) Sum of roots: } \alpha + (\alpha + 3) = 7$$

So $\alpha = 2$ and the roots are 2 and 5.

$$\text{Product of roots: } 10 = \frac{c}{2}$$

$$c = 20$$

QUESTION THIRTEEN

(a) (i) $P = (6 - x) + (10 - x) + (4x + 6 - x^2)$
 $= 2x - x^2 + 22$

(ii) $P' = 2 - 2x$

When $x = 1$, $P' = 0$ and $P'' = -2$.

Hence the stationary point at $x = 1$ when $P = 23$ is a maximum (since $P'' < 0$).

(iii) There are no other stationary points, but the minimum will occur at one end of the domain $0 \leq x \leq 3$. At $x = 0$, $P = 22$. At $x = 3$, $P = 19$. Hence the minimum perimeter is $P = 19$ units, when $x = 3$.

(b) In $\triangle ODA$, $\frac{DA}{OA} = \cos 30^\circ$

$$DA = 12\sqrt{3}$$

In $\triangle BOA$, $\frac{OA}{BA} = \cos 30^\circ$

$$BA = 24 \div \frac{\sqrt{3}}{2}$$

$$BA = 16\sqrt{3}$$

$$BD = BA - DA$$

$$= 4\sqrt{3}$$

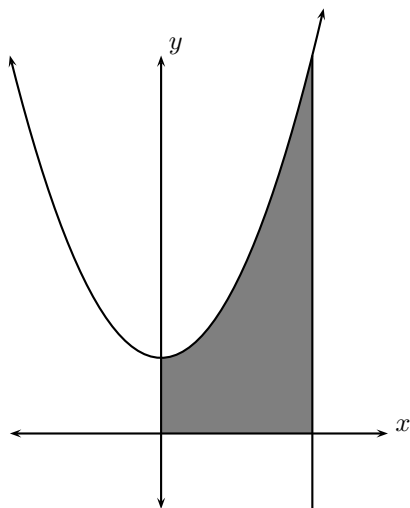
In $\triangle BCD$, $\frac{BC}{BD} = \cos 60^\circ$ (Angle sum of triangle)

$$BC = 4\sqrt{3} \times \frac{1}{2}$$

$$= 2\sqrt{3}$$

There are many ways of arriving at this result, either by trigonometry or similarity.

(c)



$$\begin{aligned} \text{Volume} &= \int_0^2 \pi y^2 dx \\ &= \int_0^2 \pi (x^2 + 1)^2 dx \\ &= \pi \times \int_0^2 (x^4 + 2x^2 + 1) dx \\ &= \pi \times \left[\frac{1}{5}x^5 + \frac{2}{3}x^3 + x \right]_0^2 \\ &= \pi \times \left(\frac{32}{5} + \frac{16}{3} + 2 \right) \\ &= \frac{206}{15}\pi \end{aligned}$$

QUESTION FOURTEEN

(a) (i) $f'(x) = 2x(16 - x^2)^8 + x^2 \times -2x \times 8(16 - x^2)^7$

(ii) $f'(x) = 2x(16 - x^2)^7(16 - x^2 - 8x^2)$
 $= 2x(4 - x)(4 + x)(16 - 9x^2)$
 $= 2x(4 - x)(4 + x)(4 - 3x)(4 + 3x)$

hence $f'(x) = 0$ when $x = 0$, $x = 4$, $x = -4$, $x = \frac{4}{3}$ or $x = -\frac{4}{3}$.

(b) (i) $T_n = ar^{n-1}$
 $= 2 \times 1.02^{n-1}$

(ii) $T_{52} = 2 \times 1.02^{51}$
 $= 5.49$

The final loaf costs \$5.49.

(iii) $4.00 < 2 \times 1.02^{n-1}$
 $1.02^{n-1} > 2$
 $n > 1 + \log_{1.02} 2$
 $n > 1 + \log 2 \div \log 1.02$
 $n > 36.003$

The price of a loaf has doubled by the start of week 37.

(iv) $S_n = a \frac{r^n - 1}{r - 1}$
 $S_{52} = 2 \frac{1.02^{52} - 1}{0.02}$

$S_{52} = 180.03$

The total cost over the 52 week period is \$180.03.

(c) For $x < 0$, the cubic lies above the line and this area is:

$$\begin{aligned} \int_{-2}^0 \left((x^3 - x^2 - 2x) - (4x) \right) dx &= \int_{-2}^0 \left(x^3 - x^2 - 6x \right) dx \\ &= \left[\frac{1}{4}x^4 - \frac{1}{3}x^3 - 3x^2 \right]_{-2}^0 \\ &= \frac{16}{3} \end{aligned}$$

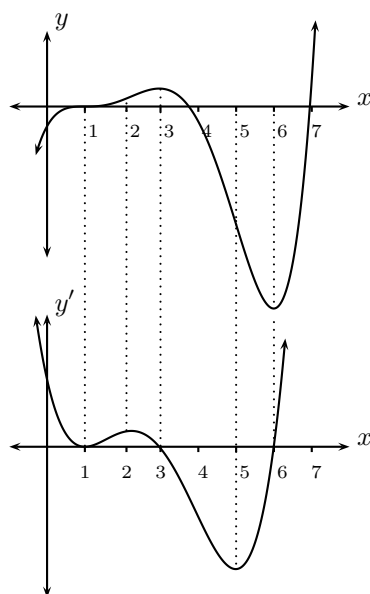
For $x > 0$, the line lies above the cubic and this area is:

$$\begin{aligned} \int_0^3 \left(4x - (x^3 - x^2 - 2x) \right) dx &= \int_0^3 \left(6x - x^3 + x^2 \right) dx \\ &= \left[3x^2 - \frac{1}{4}x^4 + \frac{1}{3}x^3 \right]_0^3 \\ &= \frac{63}{4} \end{aligned}$$

Hence the total area = $\frac{16}{3} + \frac{63}{4} = \frac{253}{12}$.

QUESTION FIFTEEN

(a)



(b) (i) There is one x -intercept: $x = -1$

(ii) The vertical asymptote is the vertical line $x = 0$.

(iii) $y = \frac{x+1}{x^2}$

$$y = \frac{\frac{1}{x} + \frac{1}{x^2}}{1}$$

$$y \rightarrow 0 \quad \text{as } x \rightarrow \pm\infty$$

Hence $y = 0$ is a horizontal asymptote.

(iv) $y = \frac{x+1}{x^2}$

$$y' = \frac{1(x^2) - (x+1)2x}{x^4}$$

$$y' = \frac{-x^2 - 2x}{x^4}$$

$$y' = \frac{-(x+2)}{x^3}$$

(v) Hence $y' = 0$ when $x = -2$.

When $x = -2$, $y = -\frac{1}{4}$

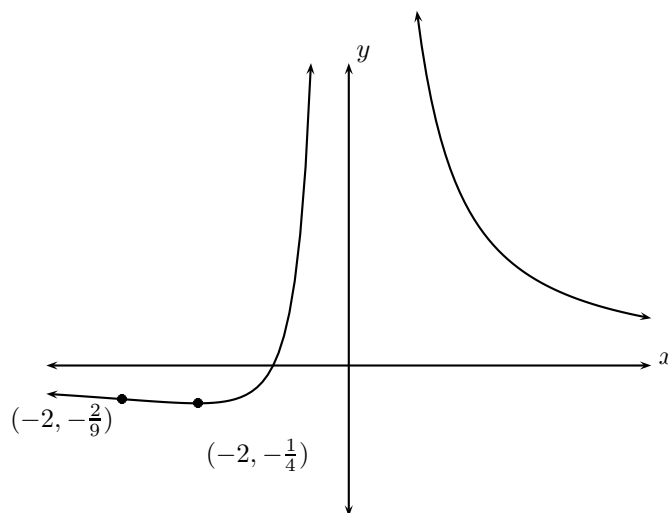
When $x = -2$, $y'' = \frac{2(-2) + 6}{(-2)^4} > 0$, hence $(-2, -\frac{1}{4})$ it is a local minimum.

(vi) There is a possible point of inflexion when $y'' = 0$, that is when $x = -3$. Testing for a change in concavity, we have:

x	-4	-3	-1
y''	$-\frac{2}{256}$	0	4
	$\cap$	$\cdot$	$\cup$

Since there is a change in concavity, $(-3, -\frac{2}{9})$ IS a point of inflexion.

(vii)

**QUESTION SIXTEEN**

- (a) Intersecting the cubic
- $y = x^3 + 2x^2 - 3x$
- and tangent
- $y = mx$
- gives

$$x^3 + 2x^2 - 3x = mx$$

$$x^3 + 2x^2 - (m+3)x = 0$$

$$x(x^2 + 2x - m - 3) = 0$$

The line $y = mx$ will be a tangent if the discriminant of the quadratic $x^2 + 2x - m - 3$ is 0. The discriminant is:

$$\Delta = 4 + 4 \times 1 \times (m+3)$$

$$= 4 \times (m+4)$$

Hence we have a repeated root and thus a tangent when $m = -4$. There will also be a tangent when $m = -3$, because $x(x^2 + 2x) = 0$ has a repeated root $x = 0$.

Thus the two tangents are $y = -3x$ and $y = -4x$.

$$(b) (i) \frac{d}{dx}(x^3 \ln x) = 3x^2 \ln x + x^3 \times \frac{1}{x}$$

$$\frac{d}{dx}(x^3 \ln x) = 3x^2 \ln x + x^2$$

$$(ii) \frac{d}{dx}(x^3 \ln x) = 3x^2 \ln x + x^2$$

Hence,

$$x^3 \ln x = \int 3x^2 \ln x \, dx + \frac{1}{3}x^3$$

$$\int 3x^2 \ln x \, dx = x^3 \ln x - \frac{1}{3}x^3$$

$$\int x^2 \ln x \, dx = \frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 + C$$

$$\begin{aligned} (iii) \int_0^1 x^2 \ln x \, dx &= \lim_{\epsilon \rightarrow 0} \int_{\epsilon}^1 x^2 \ln x \, dx \\ &= \left(-\frac{1}{3}(1)^3 \ln 1 - \frac{1}{9}(1)^3\right) - \lim_{\epsilon \rightarrow 0} \left(-\frac{1}{3}(\epsilon)^3 \ln \epsilon - \frac{1}{9}(\epsilon)^3\right) \\ &= \left(0 - \frac{1}{9}\right) - (0 - 0) \\ &= -\frac{1}{9} \end{aligned}$$

To justify this limit we have used the fact that ϵ^3 dominates $\ln \epsilon$ as $\epsilon \rightarrow 0$.

- (c) (i) As in the diagram on the paper, set up a coordinate system with origin
- O
- at the eye of the observer. Then

$$BA = b - a$$

$$AE = \sqrt{a^2 + x^2} \quad \text{By Pythagoras in } \triangle AOE$$

$$BE = \sqrt{b^2 + x^2} \quad \text{By Pythagoras in } \triangle BOE$$

Hence by the cosine rule in $\triangle ABE$,

$$\begin{aligned}\cos \theta &= \frac{AE^2 + BE^2 - BA^2}{2AE \times BE} \\&= \frac{(a^2 + x^2) + (b^2 + x^2) - (b - a)^2}{2\sqrt{a^2 + x^2}\sqrt{b^2 + x^2}} \\&= \frac{a^2 + x^2 + b^2 + x^2 - b^2 - a^2 + 2ab}{2\sqrt{a^2 + x^2}\sqrt{b^2 + x^2}} \\&= \frac{2x^2 + 2ab}{2\sqrt{a^2 + x^2}\sqrt{b^2 + x^2}} \\&= \frac{x^2 + ab}{\sqrt{a^2 + x^2}\sqrt{b^2 + x^2}} \\(ii) \tan^2 \theta &= \frac{1 - \cos^2 \theta}{\cos^2 \theta} \\&= \left(1 - \frac{(x^2 + ab)^2}{(a^2 + x^2)(b^2 + x^2)}\right) \times \frac{(a^2 + x^2)(b^2 + x^2)}{(x^2 + ab)^2} \\&= ((a^2 + x^2)(b^2 + x^2) - (x^2 + ab)^2) \times \frac{1}{(x^2 + ab)^2} \\&= (a^2b^2 + a^2x^2 + b^2x^2 + x^4 - (x^4 + a^2b^2 + 2x^2ab)) \times \frac{1}{(x^2 + ab)^2} \\&= (a^2x^2 + b^2x^2 - 2x^2ab) \times \frac{1}{(x^2 + ab)^2} \\&= \frac{x^2(a^2 + b^2 - 2ab)}{(x^2 + ab)^2} \\&= \frac{x^2(a - b)^2}{(x^2 + ab)^2}\end{aligned}$$

Now since $b > a$ and $\tan \theta > 0$, we have:

$$\tan \theta = \frac{x(b - a)}{x^2 + ab}$$

(iii) Since $\tan \theta$ is an increasing function on $0^\circ \leq \theta < 90^\circ$, to maximise θ it is enough to maximise $\tan \theta$.

By the quotient rule on this expression,

$$\begin{aligned}\frac{d}{dx} \tan \theta &= \frac{d}{dx} \left(\frac{x(b - a)}{x^2 + ab} \right) \\&= \frac{(b - a)(x^2 + ab) - x(b - a)2x}{(x^2 + ab)^2} \\&= \frac{(b - a)(x^2 + ab - 2x^2)}{(x^2 + ab)^2} \\&= \frac{(b - a)(ab - x^2)}{(x^2 + ab)^2}\end{aligned}$$

This function has a stationary point when $x^2 = ab$, i.e. when $x = \sqrt{ab} > 0$.

We need to show that this point is a maximum, either using a table of signs of the derivative, or by the second derivative test.

The second derivative is:

$$\begin{aligned}\frac{d^2}{dx^2} \tan \theta &= \frac{(b - a)(-2x)(x^2 + ab)^2 - (b - a)(ab - x^2)2(x^2 + ab)(2x)}{(x^2 + ab)^4} \\&= \frac{-2(b - a)x(x^2 + ab)(x^2 + ab - 2(ab - x^2))}{(x^2 + ab)^4} \\&= \frac{-2(b - a)x(x^2 + ab)(3x^2 - ab)}{(x^2 + ab)^4} \\&= \frac{-2(b - a)\sqrt{ab}(2ab)(2ab)}{(2ab)^4} \quad \text{when } x^2 = ab \\&< 0\end{aligned}$$

Hence we have a local maximum when $x = \sqrt{ab}$.

If we choose instead to bracket the zero of the derivative in a table of signs, obvious values to test are $x = 0$ and $x = 2\sqrt{ab}$.

x	0	$\sqrt{ab}$	$2\sqrt{ab}$
$\frac{d}{dx} \tan \theta$	$\frac{(b-a)(ab)}{(ab)^2}$	0	$\frac{(b-a)(-3ab)}{(3ab)^2}$
sign	+	0	-

Hence again we see that $x = \sqrt{ab}$ is a local maximum.

BDD